```
str_1 = "Use \t this formula when \n your data includes \\ every single item or person you want to \'study\'"
print(str_1)
```

```
Use      this formula when
 your data includes \ every single item or person you want to 'study'
```

```
zaheela = 65
sarah = 75
atif = 85
abdullah = 90
print(zaheela)
print(abdullah)
```

```
65
90
```

```
#list: To store multiple data in one variable.List are mutable(can be change)
#list used square brakets [] #dictionary
marks = [65,75,85,90]
marks
```

```
[65, 75, 85, 90]
```

```
bio_data = ["uzair","shakir",35,3.9,True,None] #Datatypes(5 types)
#int,str,float,boolean,none
bio_data
```

```
['uzair', 'shakir', 35, 3.9, True, None]
```

```
type(bio_data)
```

```
list
```

```
len(bio_data)
```

```
6
```

```
#Indexing #T = 0 , h = 1, i = 2, s = 3, space = 4......
str_2 = "This is python"
str_2[0]
```

```
'T'
```

```
student = ["Osama","hassan",3.9,"A",20,True]
student[3]
```

```
'A'
```

```
student = ["Osama","hassan",3.9,"A",20,True," "] #string = " "
student[6]
```

```
' '
```

```
student = ["Osama","hassan",3.9,"A",20,True," "]
print(student)
```

```
['Osama', 'hassan', 3.9, 'A', 20, True, ' ']
```

```
a = 39
type(a)
```

```
int
```

```
a = "39"
type(a)
```

```
str
```

```
student = ["Osama","hassan",3.9,"A",20,True," "]
student[7]
```

```
---------------------------------------------------------------------------
IndexError                                Traceback (most recent call last)
/tmp/ipykernel_453/1810518954.py in <cell line: 0>()
      1 student = ["Osama","hassan",3.9,"A",20,True," "]
----> 2 student[7]

IndexError: list index out of range
```

Next steps: Explain error

```
student
```

```
['Osama', 'hassan', 3.9, 'A', 20, True, ' ']
```

```
student = ["Osama","hassan",3.9,"A",20,True," "]
student[2] = 3.5
student
```

```
['Osama', 'hassan', 3.5, 'A', 20, True, ' ']
```

```
bio_data = ["ali","Truck Driver"]
bio_data = ["muhammad","Truck","Driver"]
```

```
#slicing
bio_data = ["ali","Truck Driver",42,"matric",4.5]
bio_data[0:2] #[startind_index,ending_index(does not count)]
```

```
['ali', 'Truck Driver']
```

```
bio_data = ["ali","Truck Driver",42,"matric",4.5]
bio_data[1:]
```

```
['Truck Driver', 42, 'matric', 4.5]
```

```
bio_data[:2]
```

```
['ali', 'Truck Driver']
```

```
bio_data = ["ali","Truck Driver",42,"matric",4.5]
bio_data[-1]
bio_data[-2]
```

```
'matric'
```

```
bio_data[-3:-1]
```

```
[42, 'matric']
```

```
bio_data[-3:]
```

```
[42, 'matric', 4.5]
```

```
bio_data[-1:]#[lower_number:highest_number]
```

```
[4.5]
```

```
bio_data[-1:-3]
```

```
[]
```

```
bio_data = ["ali","Truck Driver",42,"matric",4.5]
bio_data[1:3]
```

```
['Truck Driver', 42]
```

```
#list_methods
num = [1,2,3,4,5,6,7]
num.append(8)
num
```

```
[1, 2, 3, 4, 5, 6, 7, 8]
```

```
integer = [2,1,4,5,7,6,3]
integer.sort()
integer#ascending order
```

```
[1, 2, 3, 4, 5, 6, 7]
```

```
integer.sort(reverse=True)
integer #descending order
```

```
[7, 6, 5, 4, 3, 2, 1]
```

```
integer = [2,1,4,5,7,6,3]
integer.reverse()
integer
```

```
[3, 6, 7, 5, 4, 1, 2]
```

```
fruits = ["grapes","kiwi","oranges","apple","pineapple","leechi"]
fruits.sort()
fruits
```

```
['apple', 'grapes', 'kiwi', 'leechi', 'oranges', 'pineapple']
```

```
number = [1,5,9] #insert(index,element)
number.insert(2,4)
number
```

```
[1, 5, 4, 9]
```

```
cls = [1,2,1,3,4,1] #remove(element)
cls.remove(1)
cls
```

```
[2, 1, 3, 4, 1]
```

```
name = ["uzair","shakir",1,"uzair"]
name.remove("uzair") #First occurance
name
```

```
['shakir', 1, 'uzair']
```

```
name = ["uzair","shakir",1,"uzair"]
name.pop(1) #pop(index_number)
name
```

```
['uzair', 1, 'uzair']
```

```
a = [1,2,3,4]
b = a.copy()
b.reverse()
b
```

```
[4, 3, 2, 1]
```

```
c = [3,1,2,6,5]
d = c.copy()
d.sort()
d
```

```
[1, 2, 3, 5, 6]
```

```
name = ["uzair","shakir",1,"uzair"]
name.pop(1) #pop(index_number)
name
```

```
['uzair', 1, 'uzair']
```

```
name = ["uzair","shakir",1,"uzair"]
name.count("uzair")
```

```
2
```

```
#Tuples ()#enter + shift
tup = (1,2,3)
print(tup)
type(tup)
```

```
(1, 2, 3)
tuple
```

```
tup[1] #indexing
```

```
2
```

```
tup = (1,2,"uzair",3.9)
print(tup)
```

```
(1, 2, 'uzair', 3.9)
```

```
tup
```

```
(1, 2, 'uzair', 3.9)
```

```
print((1,2,3,4))
```

```
(1, 2, 3, 4)
```

```
lst = [1,2,3,4]
lst[2] = 5
lst
```

```
[1, 2, 5, 4]
```

```
tp = (1,2,3,4)
tp[2] = 5
tp
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
/tmp/ipykernel_1319/798647329.py in <cell line: 0>()
      1 tp = (1,2,3,4)
----> 2 tp[2] = 5
      3 tp

TypeError: 'tuple' object does not support item assignment
```

Next steps: Explain error

```
tup_1 = (1)
type(tup_1)
```

```
int
```

```
Start coding or generate with AI.
```

```
tup_1 = (1,)
type(tup_1)
tup_1
```

```
(1,)
```

```
#slicing
r = (10,20,30,50,60)
r[1:4]
```

```
(20, 30, 50)
```

```
r = (10,20,30,50,60)
r[1:7]
```

```
(20, 30, 50, 60)
```

```
lst=[1,2,3,4,5,6]
lst[1:7]
```

```
[2, 3, 4, 5, 6]
```

```
r[-3:]
```

```
(30, 50, 60)
```

```
r[:-3]
```

```
(10, 20)
```

```
r = (10,20,30,50,60)
e = r[1:4]
```

```
e
```

```
(20, 30, 50)
```

```
m = (10,20,30,50,60)
m = m[1:4]
```

```
m
```

```
(20, 30, 50)
```

```
#Methods
#index
#count
chocolates = ("kitkat","bounty","perks","dairy_milk","jublee")
chocolates.index("perks")
```

```
2
```

```
chocolates.count("bounty")
```

```
1
```

```
len(chocolates)
```

```
5
```

```
#WAP to append the movies name in empty list
movies = []
movie_1 = movies.append(input("Enter first movie name: "))
movie_2 = movies.append(input("Enter second movie name: "))
movie_3 = movies.append(input("Enter third movie name: "))
print(movies)
```

```
Enter first movie name: A walk to remmeber
Enter second movie name: Final destination
Enter third movie name: Occean 11
['A walk to remmeber', 'Final destination', 'Occean 11']
```

```
t = list(m)
```

```
t
```

```
[20, 30, 50]
```

```
#dictionary #key:value(item) {}
ali = {"physics":45,"chemistry":50,"maths":46}
type(ali)
```

```
dict
```

```
y = {3.5:"uzair",6:True}
print(y)
```

```
{3.5: 'uzair', 6: True}
```

```
students = {"Ali":[30,40,50,60,70],"Uzair":[20,30,25,35,45]}
students
```

```
{'Ali': [30, 40, 50, 60, 70], 'Uzair': [20, 30, 25, 35, 45]}
```

```
#Accessing
students = {"Ali":[30,40,50,60,70],"Uzair":[20,30,25,35,45]}
students["Ali"]
```

```
[30, 40, 50, 60, 70]
```

```
students = {"Ali":[30,40,50,60,70],"Uzair":[20,30,25,35,45]}
students_m = students["Ali"] = 30
students
```

```
{'Ali': 30, 'Uzair': [20, 30, 25, 35, 45]}
```

```
students["umer"] = [12,34,56,78,90]
students
```

```
{'Ali': 30, 'Uzair': [20, 30, 25, 35, 45], 'umer': [12, 34, 56, 78, 90]}
```

Start coding or generate with AI.